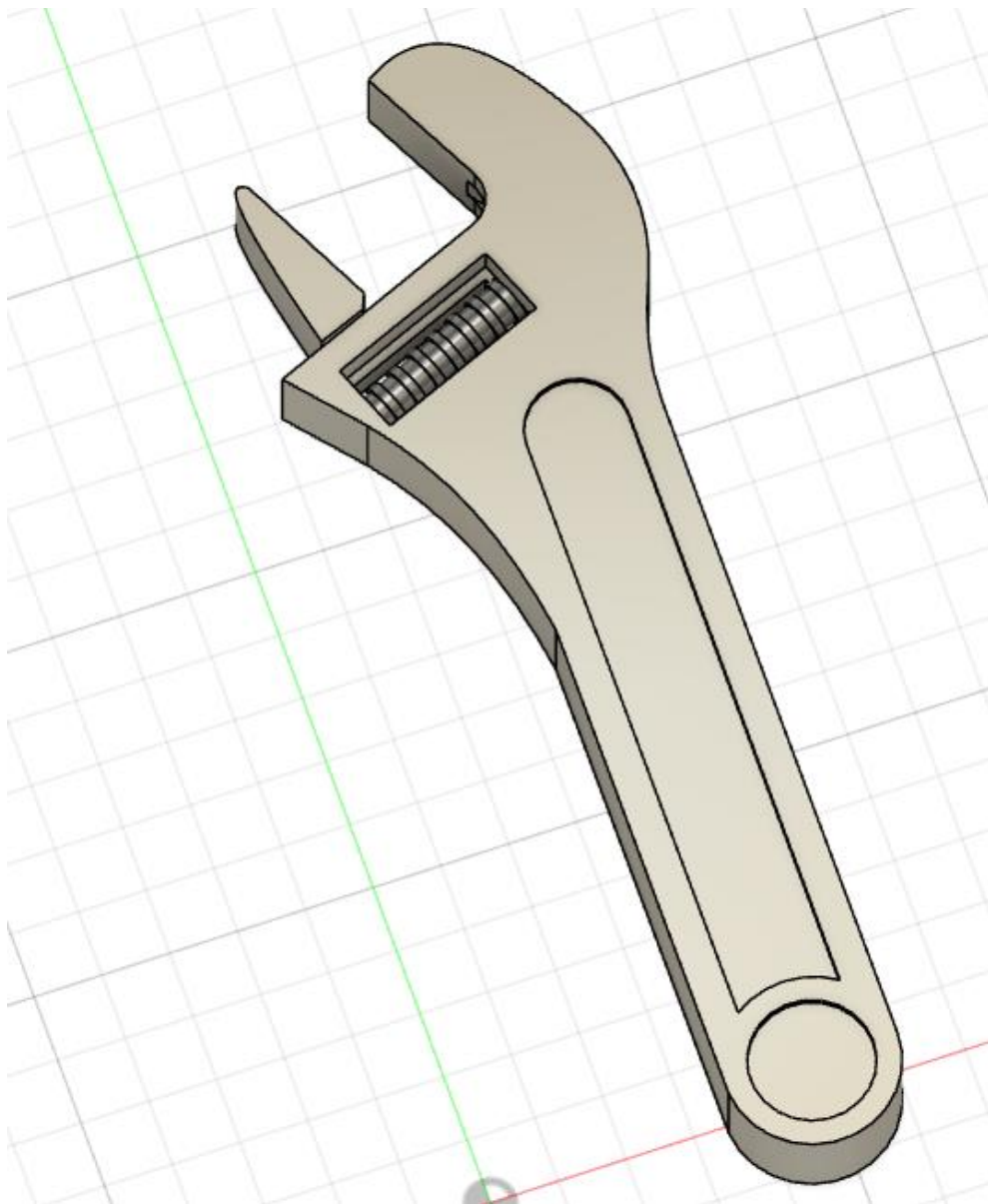
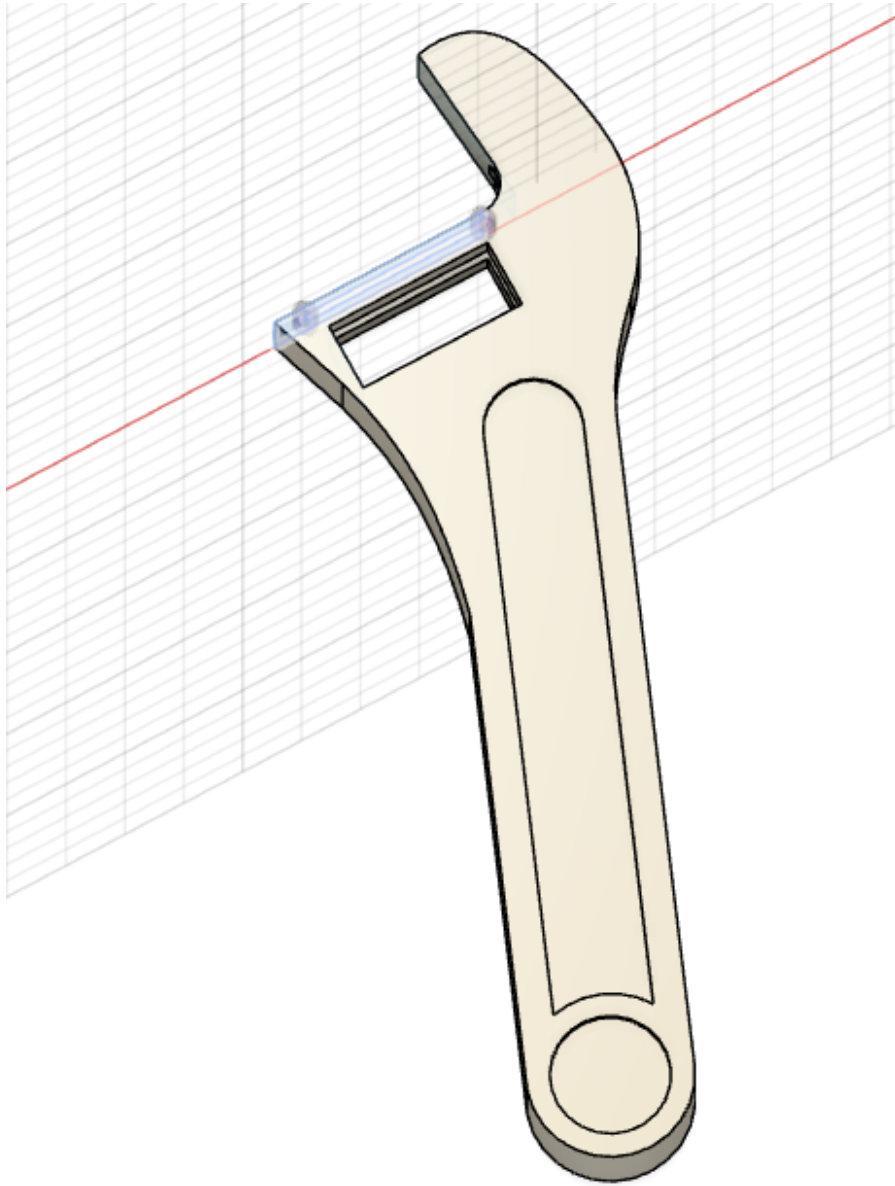


Engineering Design Workflow

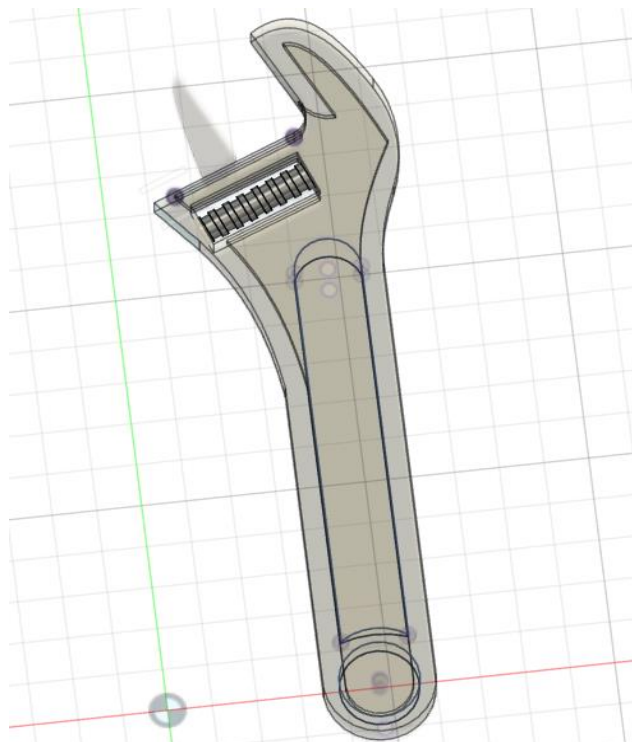
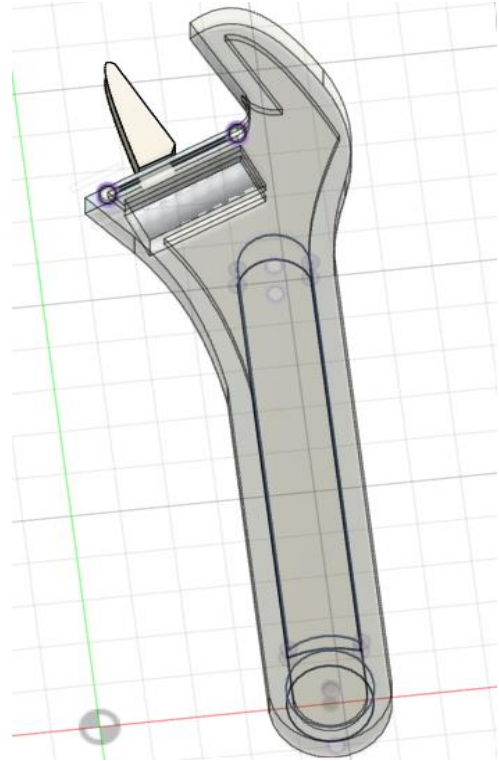
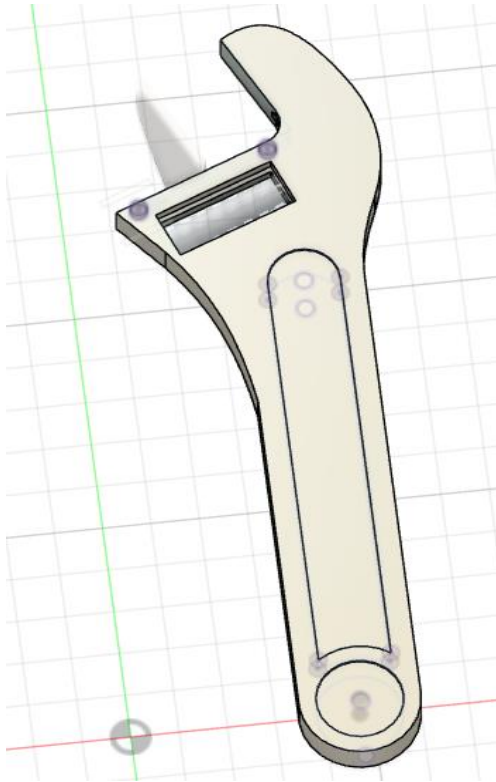
Main Image



Developed the primary structural component using parametric CAD modelling.

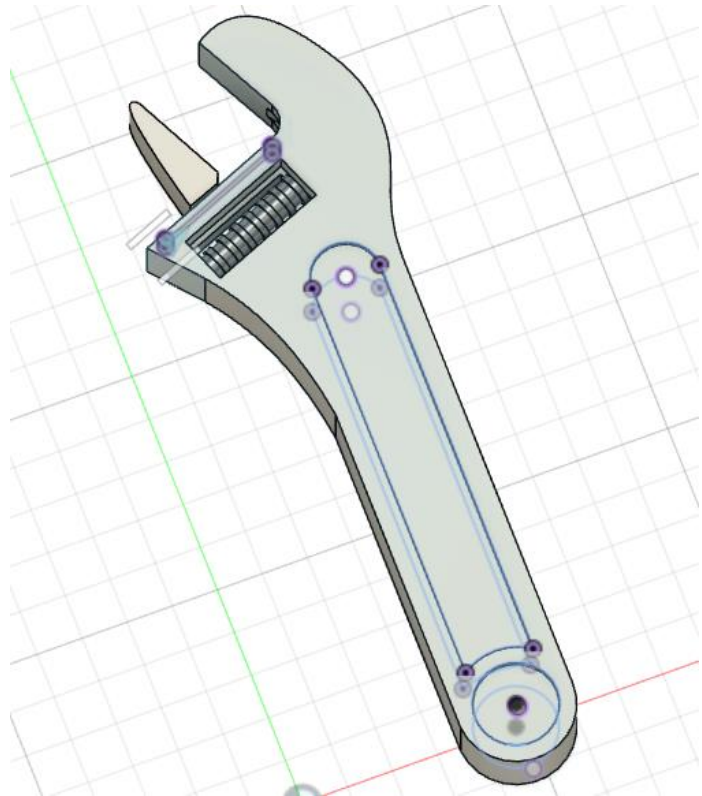
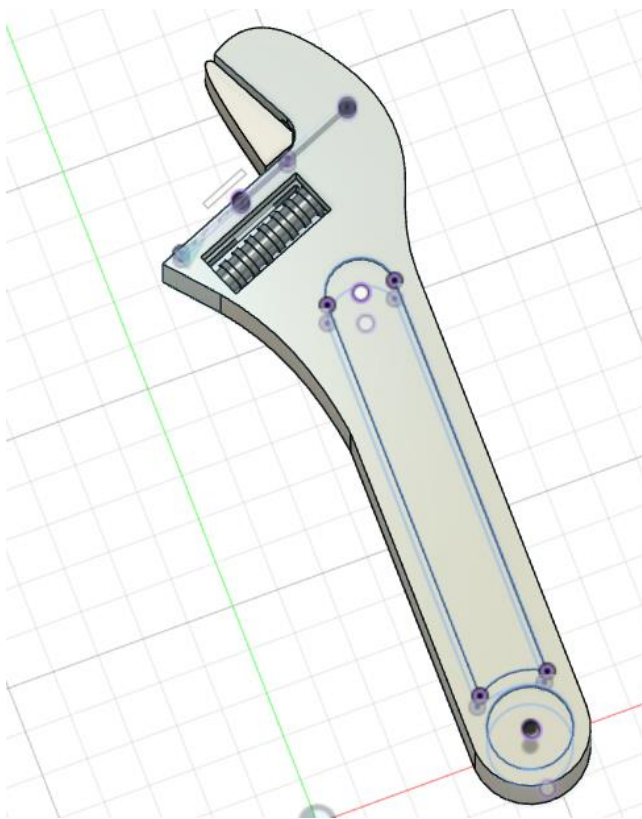


Modeled the movable jaw and adjustment screw as independent components for assembly.



Joint Movement

Assembled the components and applied motion joints to simulate jaw movement.



Engineering Challenges

Challenge	Resolution	Key Learning
Modeling the adjustment screw mechanism	Simplified the mechanism after evaluating multiple approaches.	Effective engineering balances functionality with simplicity.
Structuring the assembly components	Separated bodies into independent components.	Proper component hierarchy is essential for assemblies.
Implementing controlled jaw movement	Evaluated different joint types before selecting the most suitable configuration.	Motion simulation depends on selecting appropriate constraints.
Resolving feature failures	Corrected sketch geometry and feature dependencies.	Robust sketches reduce downstream modeling issues.
Refining the overall design	Adjusted dimensions and transitions based on design review.	Small refinements significantly improve product quality.